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Software Requirements Specification for a Collaboration Platform for Project Studies

Report for Project Study

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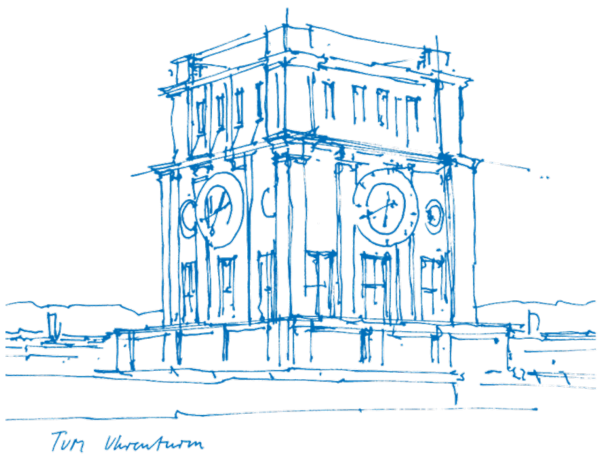

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2. Literature review

In this section we provide a review of industry and academic practices in requirements engineering, focusing on the elicitation of requirements from stakeholders. We describe the sources for our methodology, including the choice of standard, the interview-based elicitation method, and the literature on small-sample saturation. We also discuss traceability practices and prioritization schemes relevant to this project.

2.1. Requirements Engineering Grounding

2.1.1. Selection and Adaptation of the Requirements Engineering Standard

This specification follows ISO/IEC/IEEE 29148:2018 (Society, 2018) – the current international standard for requirements engineering, which supersedes IEEE 830-1998 (Society, 1998). The predecessor addressed the specification document alone, whereas 29148 covers the requirements process as a whole, including elicitation, analysis and validation. Consequently, such a form is more appropriate for this project, since much of what follows concerns how requirements were derived from stakeholder interviews and how strongly each is supported. The standard is applied in scaled-down form appropriate to a single-semester project and extended with a literature review section to meet the academic requirements of a project study report.

2.1.2. Interview-Based Requirements Elicitation

Semi-structured interviews were used to explore stakeholder needs without assuming predefined system requirements. Ferrari et al. (2022) show that interviews support an iterative elicitation process in which stakeholder initial ideas are progressively clarified and refined into requirements through the dialogue and iterative feedback. Accordingly, this project first analyzed interview data inductively to identify recurring needs and pain points, which then served as the basis for deriving platform requirements.

2.1.3. Sample Adequacy and Thematic Saturation

MacQueen et al. (1998) defined thematic saturation as the point at which no new codes emerge from additional interviews. And, the number of interviews required to reach the point of saturation is a recurring question in qualitative research.

Guest et al. (2006) found that little new information emerged after the first 12 of their 60 interviews.

Hennink et al. (2017) argue that the answer depends on what saturation is taken to mean. No new codes appeared in their data after 9 interviews, but arriving at a rich understanding of those codes took between 16 and 24. These figures come from fairly uniform groups of participants.

Hagaman & Wutich (2017) worked across 4 sites and found that 16 interviews or fewer sufficed within a single group, while themes running across groups needed 20 to 40.

These thresholds provide a basis for assessing the 26 interviews conducted for this study rather than merely reporting their number. The student group, at 13 interviews, exceeds the code-saturation threshold reported by Hennink et al. and falls within the range Hagaman and Wutich associate with thematic convergence in a single group; the pain points reported for students in Section 5.2 can therefore be regarded as reasonably stable. The company, professor and university-staff groups, at 5, 6 and 2 interviews respectively, fall below every threshold cited above, and findings specific to them are correspondingly treated as indicative. The cross-cutting findings in Section 5.5, which draw on the full sample of 26, sit within the range associated with cross-group themes, though not at its upper end. Requirements are accordingly framed throughout as informed proposals rather than validated findings.

2.1.4. Requirements Traceability

Requirements traceability is the ability to follow a requirement forward into design and implementation and backward to the source it came from. Gotel & Finkelstein (1994), drawing on interviews and questionnaires with over 100 practitioners, distinguish these two directions and argue that the backward one is where projects usually fail.

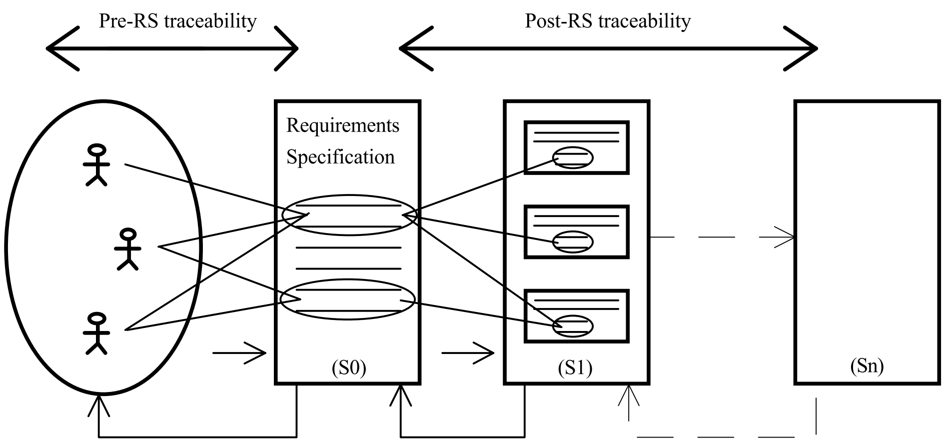

Figure 1: Illustration of the two directions of requirements traceability from Gotel & Finkelstein (1994)

Post-specification traceability, linking requirements to what was built from them, is comparatively well supported by tools. Pre-specification traceability, linking a requirement back to the stakeholder statement and rationale that produced it, tends to be neglected, and most of the problems attributed to poor traceability turn out to originate there.

Section 9 is organised along both directions: Section 9.1 provides the pre-specification record, tracing each requirement back to its pain point and interview material, while

Section 9.2 and 9.3 provide the post-specification record, mapping requirements forward to prototype screens and accounting for those left unimplemented.

2.1.5. Requirements Prioritization

Prioritization techniques differ mainly in how much they demand of the team using them. Reviewing the field, Achimugu et al. (2014) find that pairwise methods such as the Analytic Hierarchy Process become impractical as requirement counts grow, and that cost-value approaches depend on effort and benefit estimates unavailable at this stage of a project. MoSCoW asks for neither, sorting requirements into four ordinal classes on judgement alone, which suits a small team working to a fixed deadline on requirements that have not been validated. The criteria used for that sorting are set out in Section 6.1.

2.2. Academia-Industry Collaboration

2.2.1. Collaboration friction as an established phenomenon

Collaboration between universities and companies has been studied extensively, and the difficulties it produces are documented rather than incidental. Ankrah & AL-Tabbaa (2015) synthesise a fragmented body of work into a process framework covering why the two sides collaborate, the forms collaboration takes, and the organisational obstacles that recur across national contexts. Coordination and communication problems appear throughout that literature, independent of country, discipline and collaboration format. Since this problem is documented across many institutions, the difficulties reported in Section 5 are treated as a specific case of it rather than as a local peculiarity.

2.2.2. Barriers and information asymmetry between stakeholder groups

Bruneel et al. (2010) separate two kinds of barrier to university-industry collaboration. Orientation-related barriers arise from differences in what each side is trying to achieve and on what timescale. Transaction-related barriers arise from the mechanics of working together, including administration and intellectual property. Prior collaboration experience reduces the first kind, while trust between the parties reduces both. Difficulties such as identifying the right contact person or establishing what expertise is available fall into the second category, which is where a platform can plausibly intervene. This distinction is used in Section 5.5 to organise the pain points that recur across stakeholder groups.

2.2.3. Divergent motivations of academic and industry partners

The two sides enter collaboration for different reasons. Perkmann et al. (2013) show that academic engagement is driven largely by research-related motives, including access to data, funding and problems worth studying, while firms pursue capability, talent and commercial application. Neither set of motives is illegitimate, but they are not automatically aligned, and the semester rhythm that governs university work rarely matches company project timelines. A platform mediating between the two therefore cannot assume a shared objective; it has to make each side's constraints legible to the other. Section 5.5 returns to this point where the mismatch surfaced in the interview data.

2.2.4. Platforms and intermediaries as a category of solution

Where two parties struggle to find and evaluate each other, a third party can lower the cost of doing so. Howells (2006) develops a typology of innovation intermediaries and defines them as bodies acting as brokers between parties in the innovation process, performing functions such as scanning, matchmaking and coordination. The platform specified in this document sits in that category, with the difference that it serves three groups rather than two.

3. Research Approach and Limitations

3.1. Interview Methodology

The study followed a sequential design. The requirements engineering literature reviewed in Section 2.1 informed the interview guide, and a scoping discussion with BirdVision fixed what the client needed from the study. Stakeholder groups were then defined, participants recruited and interviews conducted. Responses were consolidated into a single matrix, analysed by group, and translated into requirements, from which the prototype, the traceability table, the use case diagrams and the key user flows were produced.

Interviews were semi-structured, using a common guide across all groups so that responses stayed comparable, with follow-up questions free to depart from it wherever a participant raised something unanticipated. The guide covered four areas:

1. **Current collaboration experience** — what forms of collaboration the participant had taken part in, and how they found them.
2. **Motivation** — what makes collaboration worth entering, and what makes a project attractive.
3. **Problems and barriers** — what goes wrong during collaboration, and what prevents people from engaging at all.
4. **Platform and solution needs** — asked last, so that earlier answers were not framed around a software solution.

Only the last area concerns the platform directly. The first three establish how collaboration currently works and why it is difficult, and are reported in Section 5 independently of the specification.

Four stakeholder groups were distinguished: students, companies, professors and university staff. Staff were separated from professors because their role is administrative rather than academic, and their pain points proved to differ accordingly. Participants were recruited through campus contacts and, for most company representatives, at Career Factory; Section 3.2 discusses the resulting composition and limitations.

Responses were recorded in a matrix with participants as columns and the four areas as rows, allowing each theme to be read across all participants and across groups. Section 3.3 describes how the requirements were derived from it.

3.2. Sample Composition and Limitations

Twenty-six semi-structured interviews were conducted across four stakeholder groups. Appendix B lists all participants; the distribution is summarised below.

Group	n	Institutional background
Students	13	9 TUM Heilbronn, 2 TUM Garching, 1 RWTH Aachen, 1 Reutlingen
Companies	5	Startup, industrial manufacturing, software, energy management, IT services
Professors and researchers	6	2 TUM Heilbronn, 1 TUM Munich, 1 RWTH Aachen, 1 Koblenz, 1 Passau
University staff	2	2 TUM Heilbronn

Table 1: Interview sample by stakeholder group

The sample spans five universities. Eight of the 21 university-affiliated participants are based outside TUM Campus Heilbronn, so where the same difficulties recur across institutions, they are unlikely to be local artefacts.

Two limitations shape how the company findings should be read. Most company representatives were recruited at Career Factory, a university–company event on the Bildungscampus, so the sample includes no firms that have never collaborated with a university or have stopped doing so. More importantly, four of the five hold roles in human resources, apprenticeship management or campus relations. Their interest in the university is oriented towards recruiting, and the requirements derived from this group reflect that. Needs around research cooperation, confidentiality and intellectual property are under-represented. The founder interviewed as C1 offers a partial counterweight, having neither a recruiting function nor an employer-branding budget, and is treated in Section 6 as a contrasting case rather than as one observation among five.

The company, professor and staff groups are small, and findings specific to them are treated as indicative. The staff group is smallest at two, though both are the administrators who currently run the project-study process at TUM Campus Heilbronn, so the group covers the relevant population rather than a sample of it.

This background of existing engagement is unevenly distributed across the sample. Several students and professors interviewed had no established route into university–industry collaboration, so their accounts include the perspective of those outside it. All five company representatives, by contrast, were already engaged with universities at the time of the interview. The company data therefore describes how engagement works for firms that have already committed to it, and says little about what would bring in a firm that has not.

3.3. From Interview Insights to Requirements (how findings were translated, requirements framed as informed proposals, not validated findings)

4. Overall Description

4.1. Product Perspective

4.2. Stakeholder Groups, Roles and Motivations (students, professors/supervisors, companies)

4.3. Assumptions and Constraints

4.4. Scope Boundary: TUM-Specific Focus (explicit justification: administrative/collaboration processes differ significantly across universities, so single-institution depth was prioritized over shallow generality)

4.5. Transferability to Other Institutions (brief discussion of what would need to adapt for other universities)

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6.3. [Functional Area 2]

6.4. [Functional Area 3]

6.5. [Functional Area 4]

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9.2. Prototype Coverage Table: Screen → Requirement ID

9.3. Pain Points Without Requirements and Requirements Without Prototype Coverage (brief justification)

10. Assumptions, Risks, and Open Issues

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